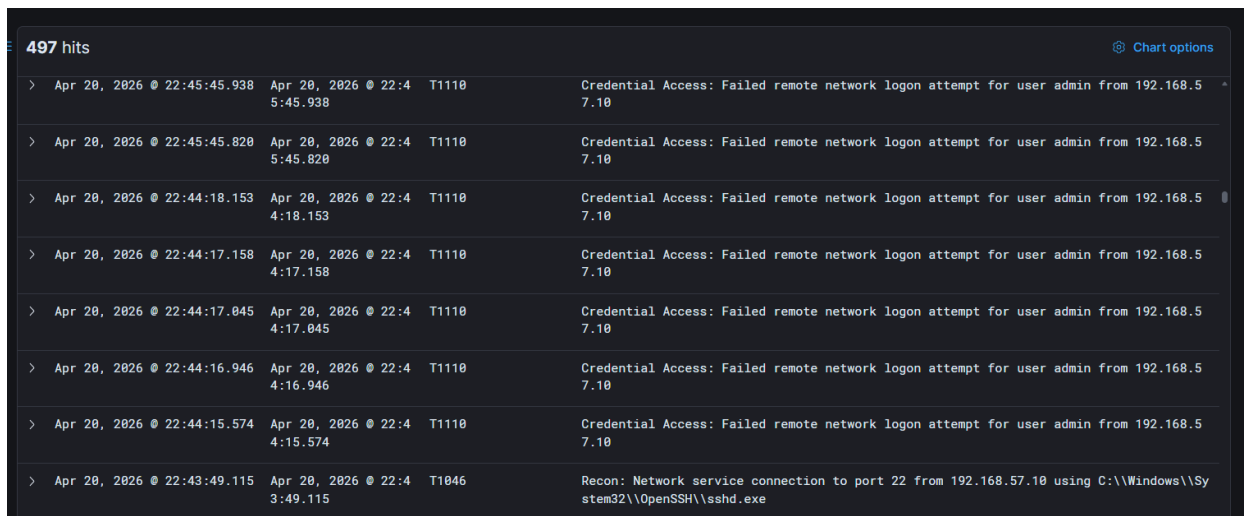


System Testing

System testing involves verifying the complete system to ensure all components work together as expected. It includes testing functionality, performance, security, and usability to identify any issues before deployment.

Test Case 1

System Testing	Objectives
Objective	To verify that the system detects and correlates multi-stage activities across different MITRE ATT&CK techniques
Action	Perform a sequence of activities that generate multiple security events and observe the system logs
Expected Result	The system logs display multiple correlated events mapped to MITRE ATT&CK techniques representing different stages of activity
Actual Result	The system successfully detected and logged multiple correlated events mapped to various MITRE ATT&CK techniques across different stages
Conclusion	Test successful



497 hits				Chart options
>	Apr 20, 2026 @ 22:45:45.938	Apr 20, 2026 @ 22:45:45.938	T1110	Credential Access: Failed remote network logon attempt for user admin from 192.168.57.10
>	Apr 20, 2026 @ 22:45:45.820	Apr 20, 2026 @ 22:45:45.820	T1110	Credential Access: Failed remote network logon attempt for user admin from 192.168.57.10
>	Apr 20, 2026 @ 22:44:18.153	Apr 20, 2026 @ 22:44:18.153	T1110	Credential Access: Failed remote network logon attempt for user admin from 192.168.57.10
>	Apr 20, 2026 @ 22:44:17.158	Apr 20, 2026 @ 22:44:17.158	T1110	Credential Access: Failed remote network logon attempt for user admin from 192.168.57.10
>	Apr 20, 2026 @ 22:44:17.045	Apr 20, 2026 @ 22:44:17.045	T1110	Credential Access: Failed remote network logon attempt for user admin from 192.168.57.10
>	Apr 20, 2026 @ 22:44:16.946	Apr 20, 2026 @ 22:44:16.946	T1110	Credential Access: Failed remote network logon attempt for user admin from 192.168.57.10
>	Apr 20, 2026 @ 22:44:15.574	Apr 20, 2026 @ 22:44:15.574	T1110	Credential Access: Failed remote network logon attempt for user admin from 192.168.57.10
>	Apr 20, 2026 @ 22:43:49.115	Apr 20, 2026 @ 22:43:49.115	T1046	Recon: Network service connection to port 22 from 192.168.57.10 using C:\\Windows\\System32\\OpenSSH\\sshd.exe

Figure 1 Initial attack activities detected by the system.

497 hits				Chart options
>	Apr 20, 2026 @ 23:00:35.616	Apr 20, 2026 @ 23:00:35.616	T1098	User account changed
>	Apr 20, 2026 @ 23:00:35.169	Apr 20, 2026 @ 23:00:35.169	T1098	User account enabled or created
>	Apr 20, 2026 @ 23:00:34.858	Apr 20, 2026 @ 23:00:34.858	T1136.001	Persistence: Local user account created
>	Apr 20, 2026 @ 23:00:34.129	Apr 20, 2026 @ 23:00:34.129	T1484	Domain Users Group Changed
>	Apr 20, 2026 @ 23:00:32.815	Apr 20, 2026 @ 23:00:32.815	T1087	A net.exe account discovery command was initiated
>	Apr 20, 2026 @ 23:00:30.418	Apr 20, 2026 @ 23:00:30.418	T1059.003	Execution: net.exe or net1.exe used to create a user account
>	Apr 20, 2026 @ 22:59:02.837	Apr 20, 2026 @ 22:59:02.837	T1550.002	Lateral Movement: Successful remote NTLM Administrator logon from 192.168.57.10
>	Apr 20, 2026 @ 22:59:00.950	Apr 20, 2026 @ 22:59:00.950	T1078	Privilege Escalation: Special privileges assigned to logon session 0x103fdb4 for user Administrator

Figure 2 Multi-Stage Attack Progression

Test Case 2

System Testing	Objectives
Objective	To verify that the system triggers an automated response when the defined risk threshold is exceeded
Action	Generate sufficient activity to increase the risk score beyond the defined threshold and observe system behavior
Expected Result	The system triggers an automated response, blocks the source, and prevents further interaction
Actual Result	The system triggered an automated response at a risk score of 0.93, blocked the source IP, and subsequent connection attempts were denied
Conclusion	Test successful

```

(kali@kali)-[~]
$ impacket-wmiexec administrator@192.168.1.102

Impacket v0.11.0 - Copyright 2023 Fortra

Password:
[*] SMBv3.0 dialect used
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out
[-] Could not connect: timed out

```

→ Connection Blocked After Automated Response

Figure 3 Connection attempts from the attacker fail after automated blocking, confirming that access has been successfully denied.

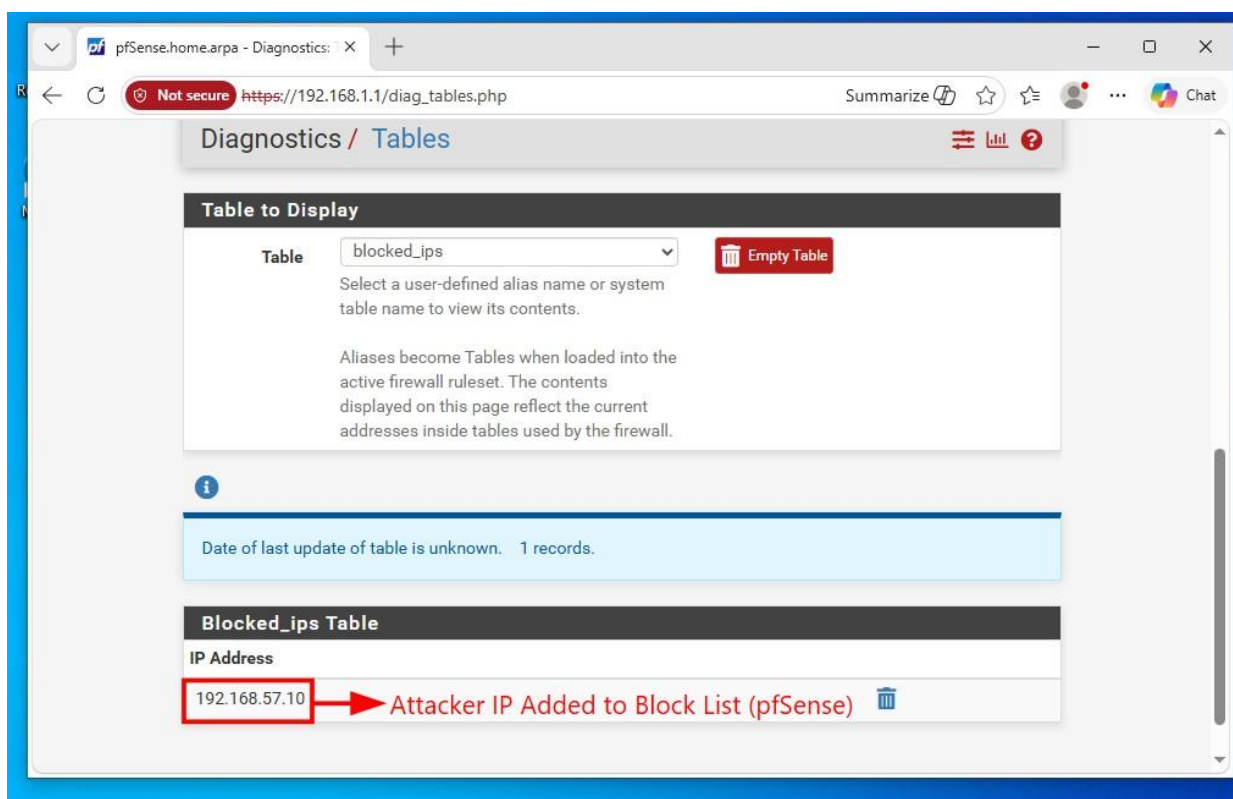


Figure 4 Attacker IP successfully added to the pfSense block table, confirming automated response enforcement.

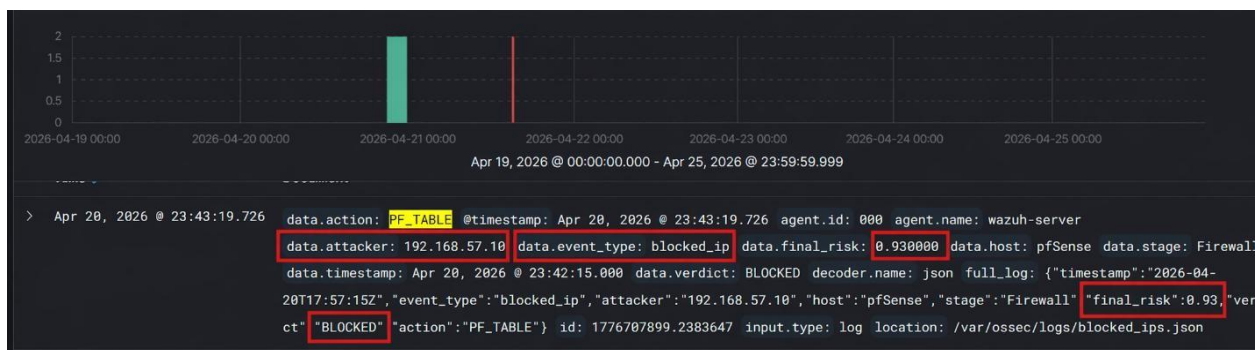


Figure 5 Security log showing blocked IP event with risk score exceeding threshold.

Test Case 3

System Testing	Objectives
Objective	To verify that the system provides clear monitoring and visibility of detected events and response actions
Action	Observe the monitoring dashboard after detection and response have occurred
Expected Result	The dashboard displays updated metrics including elevated risk score, high-risk events, and blocked source information
Actual Result	The dashboard displayed a 93% threat score, high-risk events, and the blocked source IP, confirming system visibility
Conclusion	Test successful

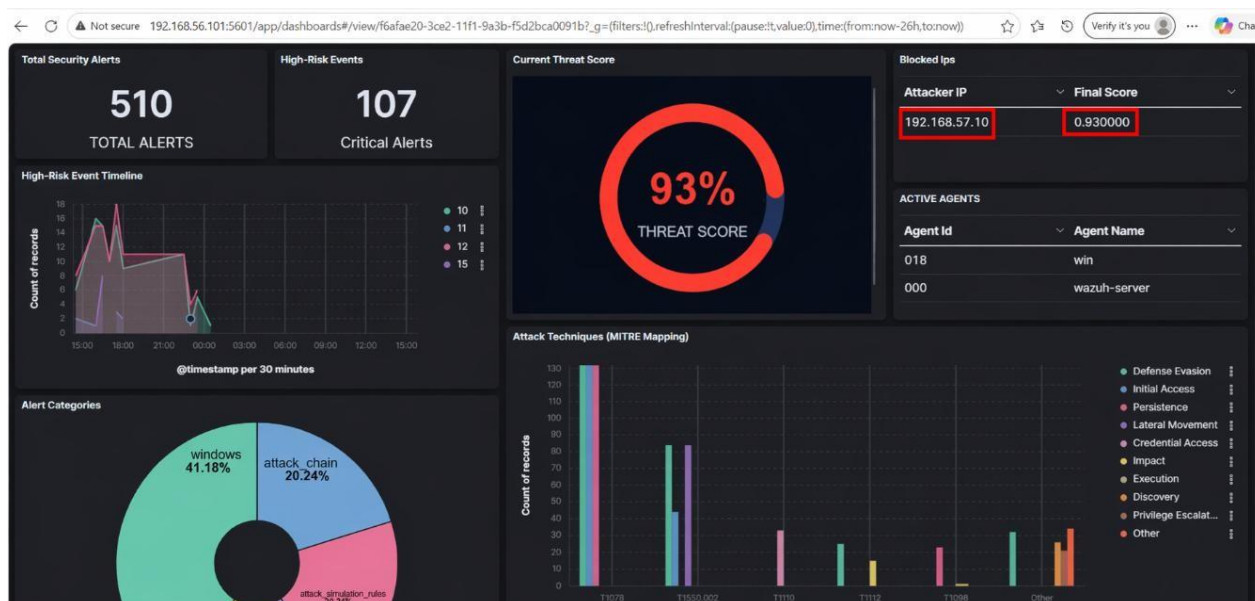


Figure 6 Dashboard displaying elevated threat score (93%) and blocked attacker IP.